Six-Arm Results, Explained

FraudNet-Synth · 5 seeds (42–46) · same global test set for every arm (95 frauds)

The three findings in one box

1. Working together (federation) clearly helps, and it helps the poorest bank the most. Bank D's fraud F1 rises from 0.456 alone to 0.796 in the federation.

2. Federated learning is about as good as pooling all data, without sharing any data. Same F1 (difference unclear); pooling is slightly better only on PR-AUC (by 0.011).

3. Synthetic data gave no reliable improvement. 33 of 36 F1/PR-AUC comparisons are "no clear difference". We report this honestly.

1. What the six arms are

We train the same classifier in six different ways and compare them. Two questions are crossed: **how do the banks work?** (alone, federated, pooled) and **do they add synthetic fraud rows?**

	Real data only	Real + validated synthetic
Isolated (each bank alone)	Arm 1: lower bound	Arm 2: augmentation without federation
Federated (Flower / FedAvg)	Arm 3: federation without augmentation	Arm 4: full FraudNet-Synth
Centralized (all data pooled)	Arm 5: upper bound (illegal in practice)	Arm 6: pooled + augmentation

Which comparison answers which question:

- **Arm 3 vs Arm 1:** does working together beat working alone?
- **Arm 2 vs 1, Arm 4 vs 3, Arm 6 vs 5:** does synthetic data help?
- **Arm 3 vs Arm 5:** how much do we lose by keeping data private instead of pooling it?

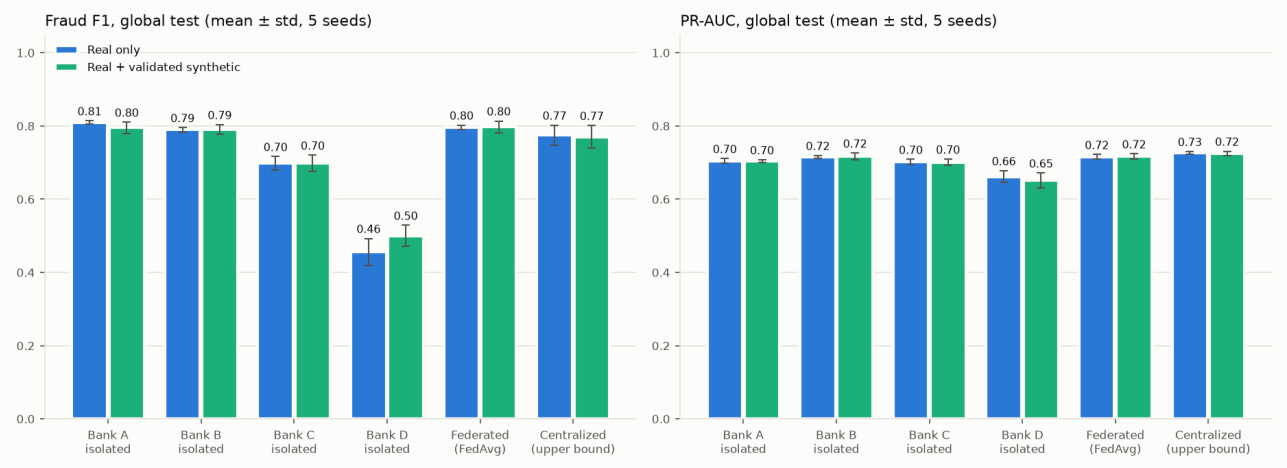
2. How to read the numbers

- **Precision:** of the payments we flagged as fraud, how many really were fraud? (Low = blocked honest customers.)
- **Recall:** of all real frauds, how many did we catch? (Low = the bank loses money.)
- **F1 (fraud):** one score that balances precision and recall. **Our headline metric.**
- **PR-AUC:** how well the model ranks frauds above genuine payments, across all thresholds.
- **ROC-AUC and Accuracy:** secondary. Accuracy is about 0.999 for *every* model, even Bank D's weak one, because 99.83% of payments are genuine. That is why accuracy is useless here.
- **"0.8093 ± 0.0048":** the mean over 5 runs with different random seeds, ± how much it varied. A small ± means a stable result.
- **Every arm is tested on the same 95 real frauds** in the global test set. Synthetic rows never enter any test set.

3. The full six-arm table (global test set, mean ± std over 5 seeds)

Arm	Model	Precision	Recall	F1 (fraud)	PR-AUC	Accuracy
1 · Isolated, real	Bank A	0.8551 ± 0.0131	0.7684 ± 0.0129	**0.8093 ± 0.0048**	0.7045 ± 0.0064	0.9994
1 · Isolated, real	Bank B	0.8936 ± 0.0012	0.7074 ± 0.0088	**0.7896 ± 0.0060**	0.7150 ± 0.0044	0.9994
1 · Isolated, real	Bank C	0.8661 ± 0.0125	0.5853 ± 0.0231	**0.6983 ± 0.0181**	0.7020 ± 0.0068	0.9992
1 · Isolated, real	Bank D	0.7994 ± 0.0103	0.3200 ± 0.0346	**0.4563 ± 0.0366**	0.6617 ± 0.0161	0.9987
1 · Isolated, real	avg of 4	0.8535 ± 0.0042	0.5953 ± 0.0092	**0.6884 ± 0.0109**	0.6958 ± 0.0068	0.9992
2 · Isolated, aug	Bank A	0.8179 ± 0.0405	0.7747 ± 0.0094	**0.7952 ± 0.0154**	0.7041 ± 0.0036	0.9993
2 · Isolated, aug	Bank B	0.8939 ± 0.0025	0.7095 ± 0.0191	**0.7910 ± 0.0129**	0.7164 ± 0.0095	0.9994
2 · Isolated, aug	Bank C	0.8710 ± 0.0057	0.5832 ± 0.0294	**0.6983 ± 0.0221**	0.7012 ± 0.0082	0.9992
2 · Isolated, aug	Bank D	0.8107 ± 0.0117	0.3621 ± 0.0274	**0.5003 ± 0.0285**	0.6511 ± 0.0203	0.9988
2 · Isolated, aug	avg of 4	0.8484 ± 0.0100	0.6074 ± 0.0084	**0.6962 ± 0.0095**	0.6932 ± 0.0077	0.9992
3 · Federated, real	federated	0.8376 ± 0.0357	0.7600 ± 0.0216	**0.7961 ± 0.0063**	0.7158 ± 0.0061	0.9993
4 · Federated, aug	federated	0.8451 ± 0.0464	0.7558 ± 0.0115	**0.7973 ± 0.0165**	0.7171 ± 0.0067	0.9994
5 · Centralized, real	centralized	0.8905 ± 0.0058	0.6863 ± 0.0390	**0.7748 ± 0.0275**	0.7267 ± 0.0037	0.9993
6 · Centralized, aug	centralized	0.8536 ± 0.0453	0.7074 ± 0.0738	**0.7701 ± 0.0312**	0.7250 ± 0.0061	0.9993

Source: results/six_arms/six_arm_global_table.csv. Red row = the weakest model (Bank D alone). Green rows = the federated models. Accuracy shown without ± (all ≈ 0.0000–0.0001).



The same numbers as bars (results/six_arms/six_arm_f1_prauc.png). Blue = real only, green = real + synthetic, whiskers = ± std over 5 seeds.

4. How we decide if a difference is real (paired 95% confidence interval)

A difference of 0.01 might just be luck. So for each comparison we compute it **seed by seed** (same seed, two arms) and build a **95% confidence interval**: the range the true difference most likely lies in.

Verdict	Rule	Meaning
▲ **higher**	whole interval above 0	a real improvement
▼ **lower**	whole interval below 0	really worse
● **no clear difference**	interval crosses 0	could be luck: we cannot claim anything

Analogy

Two students take five mock tests. If one is ahead in every test by a steady margin, the gap is real. If they swap places from test to test, you can't say who is better.

Source: results/six_arms/paired_comparisons.csv (also the "Is the difference real?" panel on the dashboard).

5. Finding 1: federation helps, most of all for the poorest bank

Arm 3 (federated) minus Arm 1 (bank alone), fraud F1 on the global test set:

Bank	Alone (Arm 1)	Federated (Arm 3)	Δ mean	95% CI	Seeds up	Verdict
Bank A (richest)	0.8093	0.7961	−0.0132	[−0.020, −0.006]	0/5	▼ lower
Bank B	0.7896	0.7961	+0.0065	[−0.007, 0.020]	4/5	● unclear
Bank C	0.6983	0.7961	+0.0978	[0.077, 0.119]	5/5	▲ higher
Bank D (poorest)	0.4563	0.7961	+0.3398	[0.299, 0.381]	5/5	▲ higher

- **Bank D gains +0.34 F1.** Alone it caught only 32% of frauds (recall 0.3200); in the federation the shared model catches 76% (recall 0.7600). This is the main success of the project.
- **Bank A loses a little (−0.013 F1).** It is the richest bank, so it has the least to learn from others. Even so, Bank A's PR-AUC went **up** in all 5 seeds (+0.0113), and its F1 on its own local test went up (+0.0444). We report the small F1 drop honestly.
- Across all 32 federation comparisons (all metrics, global and local tests): 17 higher, 3 lower, 12 unclear.

6. Finding 2: federated is about as good as centralized

Comparison	Metric	Δ mean	95% CI	Verdict
Arm 3 − Arm 5 (federated − pooled)	F1	+0.0214	[−0.017, 0.060]	● unclear
Arm 3 − Arm 5 (federated − pooled)	PR-AUC	−0.0109	[−0.021, −0.001]	▼ lower

- **F1:** no clear difference: federated (0.7961) and centralized (0.7748, which varies more: ± 0.0275) are level.
- **PR-AUC:** centralized is slightly better (0.7267 vs 0.7158), as an upper bound should be.
- **Meaning:** keeping data private costs about **0.01 PR-AUC** and no measurable F1. Pooling would be illegal in practice, so this is a very good trade.

Say **"about as good as"** centralized, never "better than".

7. Finding 3: synthetic data gave no reliable improvement

We checked augmentation in 36 comparisons on our headline metrics (F1 and PR-AUC; Arm 2 vs 1, Arm 4 vs 3, Arm 6 vs 5; per bank and overall):

Verdict	Count	Which ones
● **no clear difference**	33	almost everything
▼ **lower**	2	Bank A, federated (Arm 4 − Arm 3), own local test: F1 −0.0350, PR-AUC −0.0109
▲ **higher**	1	Bank B, centralized (Arm 6 − Arm 5), own local test: F1 +0.0524

- **Careful with Bank D:** alone, its F1 rose from 0.4563 to 0.5003 with synthetic data (+0.0440, 4 of 5 seeds). It *looks* like a gain, but the interval [−0.007, 0.095] crosses 0, so it is **not** a proven

improvement.

- **Overall federated:** Arm 4 − Arm 3 = +0.0011 F1, interval [−0.020, 0.023]: no difference.

Possible reasons (our interpretation, not proven):

- The synthetic rows are only moderately realistic (fidelity about 0.80 for CTGAN and 0.75 for the LLM). The LLM misses how columns relate, because it never saw a real row.
- We add only a 1:1 ratio of synthetic rows, a small amount of extra signal.
- Federation already solves the data-shortage problem: once banks share knowledge, extra fake rows add little.

Why this is still a good result

Augmentation was a hypothesis, not a promise. A clear "no effect" is a valid scientific result, and we report it instead of tuning the evaluation to make augmentation look good.

8. What to say to the guide (about 45 seconds)

"We compared six setups on the same 95 test frauds, over five seeds. Working alone, the smallest bank, Bank D, catches only about a third of fraud, with F1 0.456. In the federation it gets the shared model with F1 0.796, a gain of 0.34 that holds in all five seeds. The federated model is about as good as pooling all data, losing only about 0.01 PR-AUC, without any data leaving a bank. Synthetic data gave no reliable improvement: 33 of 36 comparisons were unclear, and we report that honestly. Accuracy is 99.9% for every model, so we judge by fraud F1 and PR-AUC."

9. Likely viva questions

Question	Short answer
Why not use accuracy?	It is about 0.999 for every model, even the weak one, because 99.83% of payments are genuine. It cannot tell models apart.
Why does federation lower Bank A's F1?	Bank A is the richest bank, so it gains least. Its PR-AUC and own-local F1 still went up; the poor banks gain hugely.
Is federated better than centralized?	No: about equal on F1 (difference unclear), and slightly lower on PR-AUC (−0.011). Centralized stays the upper bound.
Did synthetic data help?	No reliable effect: 33 of 36 comparisons unclear, 2 lower, 1 higher. Federation is what helped.
Why 5 seeds?	One run can be lucky. Five seeds give a mean ± std and a paired confidence interval, so we only claim differences that are consistent.